

ASUNCION, Paraguay

WMO: 862180

Lat: 25.240S

Lon: 57.519W

Elev: 89

StdP: 100.26

Time Zone: -4.00 (PRY)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.2	7.2	-0.2	3.8	12.6	1.9	4.4	11.9	11.7	19.3	10.5	19.9	1.1	180	0.396

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.6	37.2	24.1	36.2	24.2	35.2	24.2	26.9	32.9	26.5	32.3	26.1	31.8	6.0	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.3	20.7	29.6	25.0	20.3	29.4	24.3	19.4	28.8	85.6	32.4	83.5	33.2	81.3	31.4	29.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.3	9.3	8.4	DB	1.9	39.4	1.6	1.1	0.7	40.1	-0.2	40.8	-1.2	41.3	-2.3	42.1	
			WB	1.4	28.3	1.6	0.7	0.2	28.8	-0.7	29.2	-1.7	29.6	-2.8	30.2	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	23.6	28.4	27.7	26.4	23.8	19.9	18.6	18.3	20.3	22.0	24.6	25.6	27.4
	DBStd	5.32	2.14	2.53	2.77	3.78	4.11	4.92	5.45	5.65	5.13	3.82	3.06	2.59
	HDD10.0	4	0	0	0	0	0	1	3	1	0	0	0	0
	HDD18.3	240	0	0	0	6	32	59	73	47	22	2	0	0
	CDD10.0	4955	571	495	509	415	307	260	262	320	359	453	467	538
	CDD18.3	2150	313	262	251	171	80	68	73	108	132	197	217	279
	CDH23.3	21981	3672	2874	2536	1486	531	402	519	1061	1379	2023	2294	3203
Wind	WSAvg	3.6	3.3	3.2	2.9	3.1	3.1	3.5	3.8	4.2	4.3	4.2	3.8	3.5
	PrecAvg	1443	123	160	131	172	117	67	53	41	77	141	178	183
Precipitation	PrecMax	2264	261	439	275	327	371	164	167	230	201	318	491	469
	PrecMin	1052	14	31	24	34	3	0	1	0	6	21	17	43
	PrecStd	311	72	97	67	75	90	46	45	59	55	81	107	94
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.1	38.9	37.1	35.4	32.1	30.9	32.0	35.3	37.3	38.9	37.9	37.8
		MCWB	24.6	24.1	24.1	24.6	22.8	21.5	20.8	20.9	22.1	24.3	24.6	24.2
	2%	DB	36.8	36.3	35.2	33.8	30.2	29.1	30.1	33.2	35.2	36.0	35.8	36.1
		MCWB	24.5	24.6	24.5	24.5	22.7	21.4	20.6	20.8	22.0	24.1	24.4	24.7
	5%	DB	35.5	35.0	34.0	32.1	28.2	27.8	28.5	31.2	32.9	33.9	34.0	35.0
		MCWB	24.5	24.7	24.3	23.9	22.0	21.3	20.2	20.2	21.7	23.8	23.7	24.8
	10%	DB	34.1	33.2	32.2	30.2	26.2	25.8	26.2	29.1	30.2	31.8	32.1	33.6
		MCWB	24.4	24.6	23.9	23.3	21.1	20.7	19.5	19.6	20.7	23.0	23.0	24.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.5	27.5	26.8	26.7	25.1	23.5	23.3	23.4	25.1	26.6	26.8	27.7
		MCDB	33.4	33.4	32.3	32.5	29.0	27.7	28.4	31.7	32.6	33.9	33.3	33.5
	2%	WB	26.7	26.6	26.2	25.7	24.0	22.6	22.2	22.2	23.8	25.5	26.1	26.7
		MCDB	32.6	32.2	31.7	30.8	27.9	26.7	27.2	30.0	31.1	32.7	32.7	32.5
	5%	WB	26.3	26.1	25.6	24.9	22.9	21.8	21.1	21.2	22.6	24.6	25.2	26.2
		MCDB	32.0	31.7	30.9	29.7	26.8	25.8	26.1	28.5	30.1	31.5	31.3	31.8
	10%	WB	25.7	25.5	25.0	24.2	21.8	21.0	20.1	20.2	21.7	23.7	24.4	25.5
		MCDB	31.1	30.8	30.1	28.6	25.4	25.1	25.2	27.1	28.4	30.1	29.8	31.0
Mean Daily Temperature Range	5% DB	MDBR	10.6	10.2	10.3	9.9	9.1	9.0	10.2	11.4	11.3	10.5	10.8	10.6
		MCDBR	12.5	12.3	12.3	11.6	10.5	9.9	11.1	12.6	13.4	12.8	12.9	12.3
		MCWBR	3.9	3.8	4.3	4.6	5.0	4.8	5.2	5.3	5.6	5.2	5.1	4.4
	5% WB	MCDBR	10.3	10.1	10.3	10.1	9.1	8.9	10.2	11.9	11.8	11.3	10.7	10.5
		MCWBR	4.0	4.1	4.3	4.6	5.0	5.0	5.7	5.8	5.8	5.4	5.1	4.6
Clear-Sky Solar Irradiance	taub		0.411	0.413	0.401	0.400	0.380	0.403	0.397	0.505	0.586	0.504	0.414	0.417
	taud		2.403	2.405	2.419	2.400	2.418	2.346	2.348	1.995	1.802	2.087	2.352	2.373
	Ebn at Noon		934	917	898	852	824	775	798	737	722	826	924	930
	Edh at Noon		127	124	117	109	98	101	104	161	213	169	133	131
All-Sky Solar Radiation	RadAvg		6.72	6.11	5.41	4.41	3.29	2.77	3.21	3.85	4.58	5.40	6.33	6.72
	RadStd		0.44	0.45	0.39	0.45	0.45	0.28	0.32	0.27	0.49	0.57	0.38	0.41

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	+0.67	+0.31	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	N/A	+0.51	+0.20	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page